

Engineering Statistics

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Chapter 3: Probability Theory

Outline

- ▶ Sample space / events
- ▶ Set theory
- ▶ Counting events
- ▶ Probability

Introduction

- ▶ Outcome
 - ▶ The result of a random trial, process or experiment.
 - ▶ It cannot be predicted in advance with certainty.
- ▶ (Random) Experiment
 - ▶ The process of observing a phenomenon with multiple possible outcomes.
- ▶ **Sample space** ($\mathcal{S}$)
 - ▶ A set of **all** possible outcomes of an experiment.
- ▶ **Event**
 - ▶ A subset of $\mathcal{S}$.

Example: Flipping a coin

- ▶ (Random) Experiment: Flipping a coin
- ▶ Sample space: $\mathcal{S} = \{Head, Tail\}$
- ▶ Events: $\emptyset, \{Head\}, \{Tail\}, \{Head, Tail\}$
 - ▶ $\emptyset$: the empty set.

Example: Rolling a die

- ▶ Sample space: $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$
- ▶ Some examples of events:
 - ▶ The event that we observe an even number: $A = \{2, 4, 6\}$
 - ▶ The event that we observe an odd number: $B = \{1, 3, 5\}$
 - ▶ The event that we observe a number less than 5:
 $C = \{1, 2, 3, 4\}$

Example

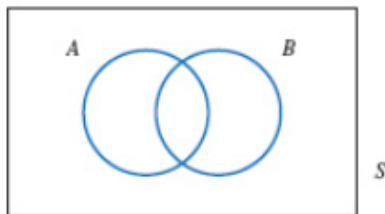
- ▶ Some sample spaces are finite or infinite.
- ▶ They can also be numerical or categorical.
 - ▶ For the weight of a component, maybe we have $\{x|x > 0\}$ or $\{x|0 < x < 150kg\}$.
 - ▶ For the blood pressure of a middle aged man $\{\text{normal, pre-hypertension, hypertension}\}$.
 - ▶ The event that a patient does not have hypertension:
 $E = \{\text{normal, pre-hypertension}\}$.

Set operation

An event is a collection of different outcomes (set). We often construct events by combining simpler events. This works the same way that set operations work in mathematics.

Venn diagram

Diagrams are often used to describe relationships between events. We can use Venn diagrams to represent a sample space and events in a sample space.



Set operators

Consider two events A and B .

- ▶ $A \cup B$: A or B (Union).
- ▶ $A \cap B$: A and B (Intersection).
- ▶ A^C : Not A . A does not occur (Complement).
- ▶ $A \subset B$: if an outcome x in A , then x is also in B (Subset).

Union of events

The **union** of two events A and B denoted $A \cup B$ is the set of outcomes that belong to A or B or both. In words, $A \cup B$ means “ A or B ”. Thus the event $A \cup B$ occurs whenever A or B (or both) occurs.

Example: Rolling a die

- ▶ The event that we observe an even number: $A = \{2, 4, 6\}$
- ▶ The event that we observe an odd number: $B = \{1, 3, 5\}$
- ▶ The event that we observe a number less than 5:
 $C = \{1, 2, 3, 4\}$

Q. What is the event that we observe an even or odd number?

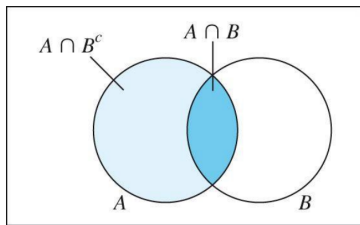
$$A \cup B = \{1, 2, 3, 4, 5, 6\} = S$$

Q. What is the event that we observe an even number or a number less than 5?

$$A \cup C = \{1, 2, 3, 4, 6\}$$

Intersection of Events

The **intersection** of two events A and B , denoted $A \cap B$ is the set of outcomes that belong to both A and B . In words, $A \cap B$ means “ A and B ”. Thus, the event $A \cap B$ occurs whenever A and B both occur.



Example: Rolling a die

Q. What is the event that we observe a number that is both even and odd?

$$A \cap B = \{\} = \emptyset$$

Q. What is the event that we observe a even number less than 5?

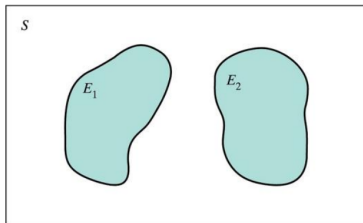
$$A \cap C = \{2, 4\}$$

Mutually Exclusive Events

Two events A and B are **mutually exclusive (disjoint)**, if they have no outcomes in common (never occur together).

$$A \cap B = \emptyset$$

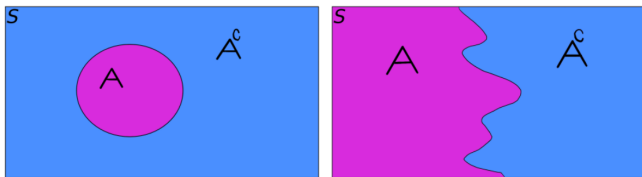
More generally, a collection of events $E_1, E_2, \dots, E_n$ are **mutually exclusive** if no two have any outcomes in common.



Venn diagram showing two disjoint (mutually exclusive) events.

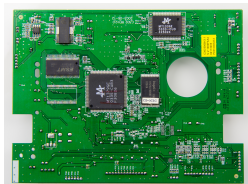
The Complement of an Event

The **complement** of an event A , denoted A^C , is the set of outcomes that do not belong to A . In words, A^C means “not A ”. Thus the event A^C occurs whenever A does not occur.



Example

A manufacturer creates Printed circuit boards (PCBs) with a thickness of 1.57mm, with a tolerance of 10%. What is the event that we do not have an acceptable thickness? Let x be the PCB thickness.



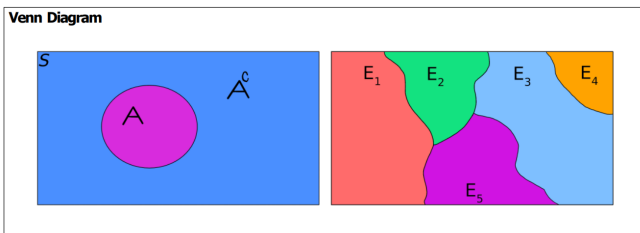
$$F = \text{acceptable} = \{x | 1.413\text{mm} \leq x \leq 1.727\text{mm}\}$$

$$F^C = \text{not acceptable} = \{x | x < 1.413\text{mm} \text{ or } x > 1.727\text{mm}\}$$

Partition of sample space

The collection of events $E_1, E_2, \dots, E_n$ is said to **partition** the sample space S if

- ▶ $E_1 \cup E_2 \cup \dots \cup E_n = S$. (exhaustive)
- ▶ $E_i \cap E_j = \emptyset, i \neq j$. (mutually exclusive)



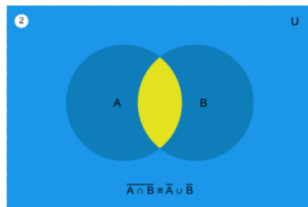
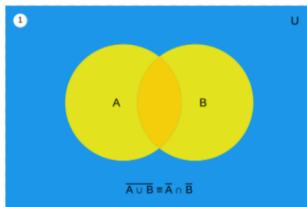
$A \cup A^c = S$, always.

The set of events $\{E_1, \dots, E_5\}$ is a partition.

De Morgan's Laws

► $(A \cup B)^C = A^C \cap B^C$

► $(A \cap B)^C = A^C \cup B^C$



Counting Events

If we want to consider the number of ways we can equip a vehicle with the options of

- ▶ Automatic or manual transmission
- ▶ With or without air conditioning
- ▶ With one of three stereo systems
- ▶ With one of four exterior colors

How many total outcomes would be in the sample space?

Counting Events

In general, if we have k stages, with n_i options at stage $i = 1, 2, \dots, k$, then the total number of possible outcomes is:

$$n_1 \times n_2 \times \cdots \times n_k$$

Example: If Virginia license plates have 3 letters A-Z followed by 4 numbers 0-9, how many distinct license plates are possible? For example,

ABC - 1234

Counting Events

Here we have $k = 7$ stages, with $n_1 = n_2 = n_3 = 26$ and $n_4 = n_5 = n_6 = n_7 = 10$, so the total number of possible license plates is:

$$26 \times 26 \times 26 \times 10 \times 10 \times 10 \times 10 = 175,760,000.$$

- ▶ Practical aside: The letter *O* likely is not used and vanity plates are possible as well.

AI Solve Example

Prompt

If Virginia license plates have 3 letters A–Z followed by 4 numbers 0–9, how many distinct license plates are possible? Explain your reasoning.

AI Output

There are 26 choices for each of the three letters and 10 choices for each of the four digits. By the multiplication principle,

$$26^3 \times 10^4 = 175,760,000.$$

Therefore, there are 175,760,000 distinct license plates.

Ordered Lists with Replacement

Suppose we will create a list of k objects, with each object selected from a collection of n possible choices. Then, we have

$$n \times n \times \cdots \times n = n^k$$

possible lists.

Example

How many binary numbers can be represented by 16 bits?

- ▶ Each bit is 0 or 1, so $n = 2$ and we are using $k = 16$ bits, so there are $2^{16} = 65,536$ possible binary numbers.

Ordered Lists without Replacement (Permutations)

Suppose we create a list of k objects, from a collection of n possible choices, **without repeating any of the previous selections**. Then, we have

$$n \times (n-1) \times (n-2) \times \cdots \times (n-k+1) = \frac{n!}{(n-k)!} = {}_n P_k$$

possible lists.

► The factorial of n :

► $n! = n \times (n-1) \times (n-2) \times \cdots \times 1$

Example

- ▶ There are 25 final presentations to be done, and we will do 5 presentations per day. How many possible lists of presentations could be constructed for the first day?

$$25 \times 24 \times 23 \times 22 \times 21 = 6,375,600$$

possible lists of presenters.

- ▶ How many possible lists of presentations could be constructed for the final day?

$$5 \times 4 \times 3 \times 2 \times 1 = 5! = 120$$

possible lists of presenters.

Unordered Lists without Replacement (Combinations)

Suppose we create a list of k objects, from a collection of n possible choices, **without repeating** any of the previous selections, and **the order DOES NOT matter**.

Unordered Lists without Replacement (Combinations)

Each one of our unordered lists would correspond to $k!$ of the ordered lists. Thus, we divide by $k!$ to obtain only the unique combinations. (Read: “ n choose k ”)

$$\frac{n!}{k!(n-k)!} = \frac{{}_nP_k}{k!} = {}_nC_k = \binom{n}{k}$$

► NOTE: $\binom{n}{k} = \binom{n}{n-k}$

Example

We have a group of 7 people, and you want to form a committee of 3 people to work on a project. The number of ways we can choose 3 people from the group of 7 to form the committee is

$$\binom{7}{3} = \frac{7!}{3!(7-3)!} = \frac{7 \times 6 \times 5}{3 \times 2 \times 1} = 35$$

So, there are 35 different combinations of selecting 3 people from the group of 7 to form the committee. Each combination represents a unique set of individuals who will work together on the project.

Example

If we want to consider the number of ways we can equip a vehicle with the options of

- ▶ Automatic or manual transmission
- ▶ With or without air conditioning
- ▶ With one of three stereo systems
- ▶ With one of four exterior colors

How many total outcomes would be in the sample space?

- ▶ $2 \times 2 \times 3 \times 4 = 48$ total options.
- ▶ $\binom{2}{1} \times \binom{2}{1} \times \binom{3}{1} \times \binom{4}{1} = 48$.

Example

Suppose we have 100 microcontrollers in a box, and unbeknownst to you, 14 of these are damaged. You collect 20 microcontrollers from the box. What is the number of possible ways that you select exactly 3 damaged microcontrollers from your collection of 20?

Example

Let's break it into pieces:

- ▶ Of the 14 damaged, how many ways can we select 3 of them?
- ▶ Of the 86 not damaged, how many ways can we select 17 of them?

$$\Rightarrow \binom{14}{3} \times \binom{86}{17}$$

Multinomial counts

You have a group of 7 people, and you want to divide them into two groups: one with 3 people and the other with the remaining 4 people.

$$\binom{7}{3} = \frac{7!}{3! \cdot 4!} = 35$$

Multinomial counts

The research and development team has 9 projects to work on. Management prefers to have three marked as top priority, four marked as medium priority, and two marked as low priority. How many different ways can you assign the projects to the priority levels?

Multinomial counts

Let's break it into pieces:

- ▶ How many ways can we arrange projects from 1st priority to 9th priority?
- ▶ How many ways could we arrange
 - ▶ the top priority projects?
 - ▶ the medium priority projects?
 - ▶ the low priority projects?

So, to remove these equivalent groupings, we divide by these group-specific factorials:

$$\frac{9!}{3! \cdot 4! \cdot 2!} = 1260.$$

Multinomial counts

We wish to assign n distinct objects each to one of k different categories, so that there will be n_i in the i th category. This implies $n_1 + n_2 + \cdots + n_k = n$. The number of ways this assignment can be made is:

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

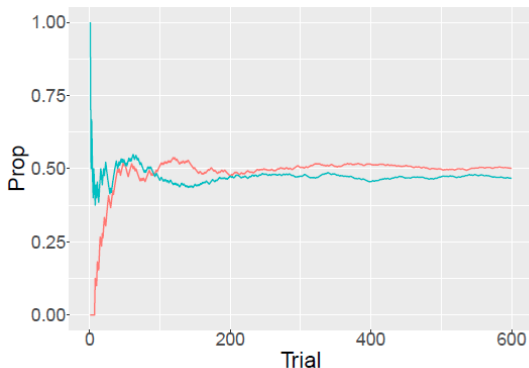
Probability

Probability is used to quantify the likelihood, or chance, that an outcome of a random experiment will occur. We denote the probability of an event E occurring by $\Pr(E)$.

How do we interpret probability?

- ▶ Relative frequency
 - ▶ The proportion of times the outcome would occur in a very long run of repetitions of the random experiment.
- ▶ Degree of belief
 - ▶ A number between 0 and 1 that expresses an individual's judgment of how likely the outcome is.
 - ▶ Any set of personal probabilities that makes sense obeys the same basic probability rules.

Example



The proportion of tosses of a coin that give a head changes as we make more tosses. Eventually, the proportion approaches to 0.5, the probability of a head.

Equally likely outcome

- ▶ For many experiments, it is reasonable to assume that all possible outcomes are **equally likely** to calculate the probability of events.
- ▶ Whenever a sample space consists of N possible outcomes that are **equally likely**, the probability of each outcome is $1/N$.

Example: Rolling a die

Assume that we have a **fair** die.

- ▶ The event that we observe an even number: $A = \{2, 4, 6\}$

$$\begin{aligned}\Pr(A) &= \Pr(\text{observe an even}\#) \\ &= \frac{\text{\#ways to see an even number}}{\text{\#ways to get a face value}} \\ &= \frac{3}{6}\end{aligned}$$

AI Example: A Brief History of Probability Theory

Prompt

Give a brief history of probability theory. Highlight the major milestones and key contributors.

AI Output

Probability theory developed from attempts to understand games of chance.

- ▶ **1654 – Pascal and Fermat:** Their correspondence about gambling problems laid the foundations of mathematical probability.
- ▶ **1657 – Huygens:** Published one of the first systematic treatments of probability.
- ▶ **1713 – Bernoulli:** *Ars Conjectandi* introduced important ideas including the law of large numbers.
- ▶ **18th–19th centuries – Bayes and Laplace:** Bayes developed ideas underlying inverse probability, while Laplace greatly expanded and systematized probability theory.
- ▶ **1933 – Kolmogorov:** Established the modern axiomatic foundation of probability using measure theory.

Probability theory has since become a mathematical foundation for statistics, engineering, science, economics, and modern data analysis.

Probability rules

1. For any event A , $0 \leq \Pr(A) \leq 1$.
2. $\Pr(S) = 1$.
3. **Addition rule**
4. **Multiplication rule**

Addition rule

- ▶ If A and B are **mutually exclusive** ($A \cap B = \emptyset$),

$$\Pr(A \cup B) = \Pr(A) + \Pr(B).$$

- ▶ $\Pr(\text{not } A) = 1 - \Pr(A).$
- ▶ For **any** two events A and B ,

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B).$$

Addition rule (disjoint)

If A and B are **mutually exclusive** ($A \cap B = \emptyset$),

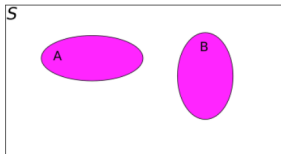
$$\Pr(A \cup B) = \Pr(A) + \Pr(B).$$

More generally, if $E_1, E_2, \dots, E_n$ are **mutually exclusive** events then

$$\begin{aligned}\Pr(\cup_{i=1}^n E_i) &= \Pr(E_1 \cup E_2 \cup \dots \cup E_n) \\ &= \Pr(E_1) + \Pr(E_2) + \dots + \Pr(E_n) \\ &= \sum_{i=1}^n \Pr(E_i).\end{aligned}$$

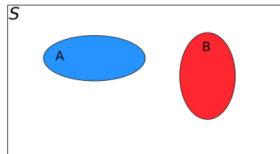
Addition rule (disjoint)

$$P(A \cup B)$$



=

$$P(A) + P(B)$$

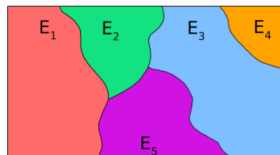


$$1 = P(S) = P(E_1 \cup E_2 \cup E_3 \cup E_4 \cup E_5)$$



=

$$P(E_1) + P(E_2) + P(E_3) + P(E_4) + P(E_5)$$



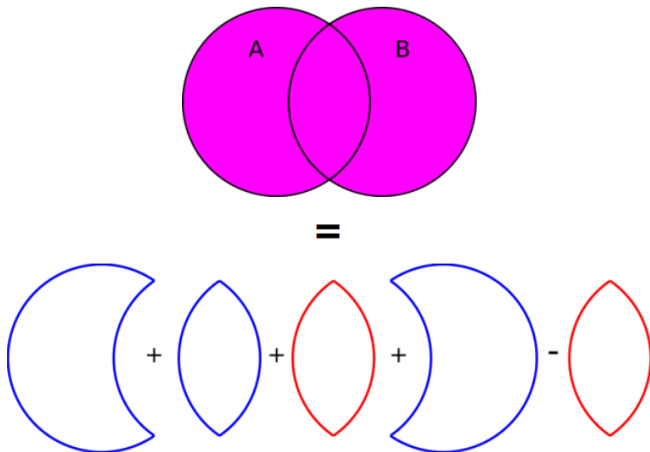
Remark

- ▶ $\Pr(A^C) = 1 - \Pr(A)$
- ▶ $\Pr(\emptyset) = 0$

Addition rule (General case)

- For **any** two events A and B ,

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B).$$



Example: Rolling a die

Assume that we have a **fair** die.

- ▶ The event that we observe an even number: $A = \{2, 4, 6\}$

$$\begin{aligned}\Pr(A) &= \Pr(2) + \Pr(4) + \Pr(6) \\ &= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} \\ &= \frac{3}{6} \\ &= \frac{1}{2}\end{aligned}$$

- ▶ What is the probability that we observe an even number or a number less than 5?

Example

According to the Registrar, 40% of students take math, 30% take chemistry, and 20% take Both.

Q. What is the probability that a randomly selected student takes either math or chemistry (or both is implied)?

$\iff$ What proportion of students take either math or chemistry or both?

$$\Pr(M \cup C) = \Pr(M) + \Pr(C) - \Pr(M \cap C) = 0.5$$

Q. What is the probability that a randomly selected student takes math but not chemistry?

$$\begin{aligned}\Pr(\text{Math but not Chemistry}) &= \Pr(M \cap C^C) \\ &= \Pr(M) - \Pr(M \cap C) \\ &= 0.2.\end{aligned}$$

Multiplication rule

- ▶ **Conditional probability**
- ▶ **Independent Events**

Conditional Probability

Consider events A and B . If event A has occurred, what is the probability that event B occurs?

- ▶ The **conditional probability** of B given A is

$$\Pr(B|A) = \frac{\Pr(A \cap B)}{\Pr(A)}, \quad \Pr(A) \neq 0.$$

Example

Q. What is the probability that a person has Brown eyes given that they have red hair?

Hair color and eye color

		Hair color			Total
		Brown	Black	Red	
Eye color	Brown	400	300	20	720
	Blue	800	200	50	1,050
	Total	1,200	500	70	1,770

Example

$$\Pr(\text{Brown eyes and Red hair}) = 20/1770$$

$$\Pr(\text{Red hair}) = 70/1770$$

$$\Pr(\text{Brown eyes}|\text{Red hair}) = \frac{\Pr(Br \cap R)}{\Pr(R)} = \frac{20/1770}{70/1770} = \frac{2}{7}.$$

Hair color and eye color

		Hair color			Total
		Brown	Black	Red	
Eye color	Brown	400	300	20	720
	Blue	800	200	50	1,050
	Total	1,200	500	70	1,770

Independent Events

Two events A and B are **independent** if knowing that one event occurs does not change the probability of the other event. That is, if

$$\Pr(A) = \Pr(A|B) \text{ or } \Pr(B) = \Pr(B|A)$$

then A and B are independent.

Multiplication Rules

For **any** two events A and B ,

$$\Pr(A \cap B) = \Pr(A) \times \Pr(B|A).$$

- ▶ If A and B are **independent**,

$$\Pr(A \cap B) = \Pr(A) \times \Pr(B).$$

Example

A boat has two motors, a main one, and a backup motor for emergencies. The probability that the main motor fails is 0.10. The probability that the backup fails (given that the main failed) is 0.05.

► **What is the probability that both motors fail?**

Let M be the event that the main motor works. $\Rightarrow M^C$ is the event that it fails. Similarly let B be the event that the backup motor works.

Example

What is the probability that both motors fail?

Let $\Pr(M^C) = 0.10$, $\Pr(B^C|M^C) = 0.05$, then

$$\Pr(M^C \cap B^C) = \Pr(M^C) \times \Pr(B^C|M^C) = (0.10)(0.05) = 0.005.$$

Checking for Independence

To check if the events A and B are independent, we see if the following are true:

1. $\Pr(A|B) = \Pr(A)$
2. $\Pr(B|A) = \Pr(B)$
3. $\Pr(A \cap B) = \Pr(A) \Pr(B)$

If **any** of the above are true, then **all** are true. We simply need to check any one (not all).

Example

Is having red hair independent of having brown eyes?

Hair color and eye color

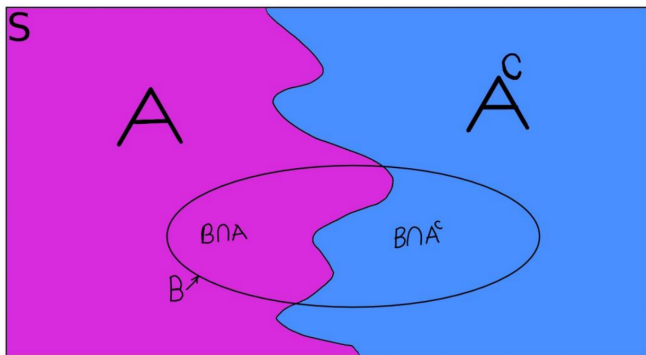
		Hair color			Total
		Brown	Black	Red	
Eye color	Brown	400	300	20	720
	Blue	800	200	50	1,050
	Total	1,200	500	70	1,770

1. $\Pr(\text{Brown eyes}|\text{Red hair}) \stackrel{?}{=} \Pr(\text{Brown eyes})$
2. $\Pr(\text{Red hair}|\text{Brown eyes}) \stackrel{?}{=} \Pr(\text{Red hair})$
3. $\Pr(\text{Brown eyes} \cap \text{Red hair}) \stackrel{?}{=} \Pr(\text{Brown eyes}) \Pr(\text{Red hair})$

Bayes rule

For any two events A and B , with $\Pr(A) \neq 0$, $\Pr(A^C) \neq 0$, and $\Pr(B) \neq 0$,

$$\Pr(B) = P\left[(A \cap B) \cup (A^C \cap B)\right] = \Pr(A \cap B) + \Pr(A^C \cap B)$$



Bayes rule

Furthermore, by the multiplication rule,

$$\Pr(B) = \Pr(A \cap B) + \Pr(A^C \cap B) = \Pr(A) \Pr(B|A) + \Pr(A^C) \Pr(B|A^C)$$

This is known as *the rule of total probability*.

Bayes rule

Bayes' rule:

$$\Pr(A|B) = \frac{\Pr(A) \Pr(B|A)}{\Pr(A) \Pr(B|A) + \Pr(A^C) \Pr(B|A^C)}.$$

Proof:

$$\begin{aligned}\Pr(A|B) &= \frac{\Pr(A \cap B)}{\Pr(B)} \\ &= \frac{\Pr(A) \Pr(B|A)}{\Pr(B)} \\ &= \frac{\Pr(A) \Pr(B|A)}{\Pr(A) \Pr(B|A) + \Pr(A^C) \Pr(B|A^C)}\end{aligned}$$

Example: Diagnosis Tests

Suppose that the proportion of people with a disease, D , is 0.08. A test is available for the diagnosis of D . If a person has D , the probability that the test shows positive is 0.95. If a person does not have D , the test shows positive with a probability of 0.1.

Q. What is the probability that a person tests positive?

Example: Diagnosis Tests

Let $+$ be the event that a person tests positive for the disease, and D be the event that the person actually has the disease.

- ▶ $\Pr(D) = 0.08 \Rightarrow \Pr(D^C) = 0.92$
- ▶ $\Pr(+|D) = 0.95 \Rightarrow \Pr(+|D^C) = 0.1$

$$\begin{aligned}\Pr(+) &= P[(+ \cap D) \cup (+ \cap D^C)] \\ &= \Pr(+ \cap D) + \Pr(+ \cap D^C) \\ &= \Pr(D) \Pr(+|D) + \Pr(D^C) \Pr(+|D^C) \\ &= 0.08 \times 0.95 + 0.92 \times 0.1 \\ &= 0.168\end{aligned}$$

Example: Diagnosis Tests

Q. If a person gets a positive test result, what is the probability that the person actually has the disease?

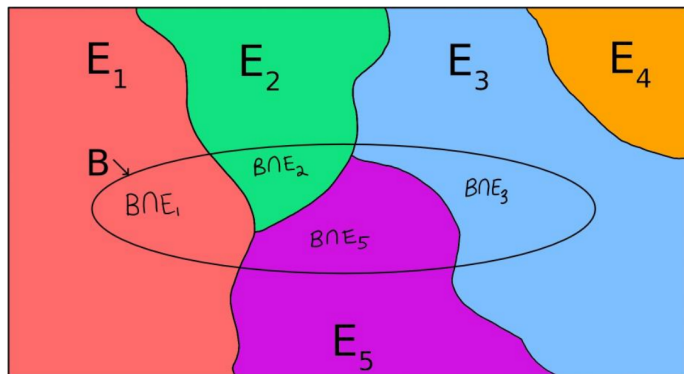
$$\begin{aligned}\Pr(D|+) &= \frac{\Pr(D \cap +)}{\Pr(+)} \\&= \frac{\Pr(D) \Pr(+|D)}{\Pr(D) \Pr(+|D) + \Pr(D^C) \Pr(+|D^C)} \\&= \frac{0.08 \times 0.95}{0.168} \\&= 0.4524.\end{aligned}$$

Bayes rule (General version)

Let $A_1, A_2, \dots, A_k$ are **mutually exclusive** events whose probabilities are not 0 and **add to exactly 1** ($A_1, A_2, \dots, A_k$ partition the sample space). Then if B is any other event,

$$\Pr(A_i|B) = \frac{\Pr(A_i) \Pr(B|A_i)}{\sum_{j=1}^k \Pr(A_j) \Pr(B|A_j)}.$$

Bayes rule (General version)



$$\begin{aligned}\Pr(B) &= \Pr(E_1 \cap B) + \Pr(E_2 \cap B) + \cdots + \Pr(E_5 \cap B) \\ &= \Pr(E_1) \Pr(B|E_1) + \Pr(E_2) \Pr(B|E_2) + \cdots + \Pr(E_5) \Pr(B|E_5)\end{aligned}$$

Example

- ▶ An underwriter studies home fires resulting from wood-burning furnaces.
- ▶ Among homes with wood-burning furnaces:
 - ▶ 30% have a Type 1 furnace,
 - ▶ 25% have a Type 2 furnace,
 - ▶ 15% have a Type 3 furnace,
 - ▶ 30% have other types.
- ▶ Over a 3-year period, the percentages resulting in fires are:
 - ▶ Type 1: 5%,
 - ▶ Type 2: 3%,
 - ▶ Type 3: 2%,
 - ▶ Other types: 4%.
- ▶ **Question:** If a fire occurs in a particular home, what is the probability that the home has a Type 1 furnace?

Example

- ▶ $\Pr(\text{Type 1}) = 0.30$, $\Pr(\text{Type 2}) = 0.25$,
 $\Pr(\text{Type 3}) = 0.15$, $\Pr(\text{Other Type}) = 0.30$.
- ▶ $\Pr(\text{Fire} | \text{Type 1}) = 0.05$, $\Pr(\text{Fire} | \text{Type 2}) = 0.03$,
 $\Pr(\text{Fire} | \text{Type 3}) = 0.02$, $\Pr(\text{Fire} | \text{Other Types}) = 0.04$.

We want to know $\Pr(\text{Type 1} | \text{Fire})$.

Example: Solution

By Bayes' rule,

$$\Pr(\text{Type 1} \mid \text{Fire}) = \frac{\Pr(\text{Fire} \mid \text{Type 1}) \Pr(\text{Type 1})}{\Pr(\text{Fire})}.$$

Using the rule of total probability,

$$\begin{aligned}\Pr(\text{Fire}) &= (0.05)(0.30) + (0.03)(0.25) \\ &\quad + (0.02)(0.15) + (0.04)(0.30) \\ &= 0.0375.\end{aligned}$$

Therefore,

$$\begin{aligned}\Pr(\text{Type 1} \mid \text{Fire}) &= \frac{(0.05)(0.30)}{0.0375} \\ &= 0.40.\end{aligned}$$

Given that a fire occurred, there is a 40% chance that the home had a Type 1 furnace.

AI Exercise: Bayes' Rule

Prompt to AI

Consider the following information about home furnaces:

- ▶ $\Pr(\text{Type 1}) = 0.30$, $\Pr(\text{Type 2}) = 0.25$, $\Pr(\text{Type 3}) = 0.15$, and $\Pr(\text{Other}) = 0.30$.
- ▶ $\Pr(\text{Fire} \mid \text{Type 1}) = 0.05$, $\Pr(\text{Fire} \mid \text{Type 2}) = 0.03$.
- ▶ $\Pr(\text{Fire} \mid \text{Type 3}) = 0.02$, $\Pr(\text{Fire} \mid \text{Other}) = 0.04$.

If a fire occurs in a particular home, what is the probability that the home has a Type 1 furnace?

Solve for $\Pr(\text{Type 1} \mid \text{Fire})$ and explain your reasoning.

Your Task

Ask an AI tool to solve the problem. Then:

- ▶ Identify the probability rule used by AI.
- ▶ Check each step of the calculation.
- ▶ Decide whether the AI's final answer and interpretation are correct.